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Building the Capacity to Conduct Joint All-Domain Operations (JADO)

Considerations for the UK

Justin Bronk and Samuel Cranny-Evans





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Contents

Abbreviations v

Executive Summary vii

Introduction **1**

What are Joint All-Domain Operations? 3

I. Where is JADO Now? **7**

II. Vignette 1: SEAD in Eastern Europe **13**

III. Vignette 2: Integrated Air and Missile Defence **19**

IV. Vignette 3: Contested Freedom of Navigation Operations **23**

Conclusion **27**

About the Authors **29**

Abbreviations

ABMS – Advanced Battle Management System

C2 – command and control

C-RAM – counter-rocket and mortar

C-UAS – counter-unmanned aerial systems

DEAD – destruction of enemy air defences

DoD – Department of Defense (US)

ELINT – electronic intelligence

FONOP – freedom of navigation operation

GMLRS – guided multiple launch rocket system

IADS – integrated air defence system

IAMD – integrated air and missile defence

IBCS – Integrated Battle Command System

ITN – Integrated Tactical Network

JADC2 – Joint All-Domain Command and Control

JADO – Joint All-Domain Operations

JLENS – Joint Land-Attack Cruise Missile Defense Elevated Netted Sensor System

LEAPP – Land Environment Air Picture Provision

MDI – multidomain integration

MESA – Multi-role Electronically Scanned Array (radar)

MEZ – missile engagement zone

NIFC-CA – Naval Integrated Fire Control – Counter Air

RCO – Rapid Capabilities Office

RPAS – remotely piloted aerial systems

SAM – surface-to-air missile

SEAD – suppression of enemy air defence systems

SIGINT – standoff signals intelligence

SLOCs – sea lines of communication

SOF – special operations forces

sUAS – small unmanned aerial systems

TLAM – Tomahawk land attack missiles

UAS – unmanned aerial systems

Executive Summary

There is a widely held quasi-philosophical position across the UK, the US and the armed forces of other nations that future wars will be decided by whichever side can best manage and share data seamlessly across all force elements. The UK Ministry of Defence (MoD) is doctrinally and conceptually committed to a vision of the future force in which seamless cross-domain connectivity and integration is the norm.

However, this high-level vision does not translate neatly into programme deliverables and prioritisation decisions. With limited resources and a host of urgent force modernisation and regeneration requirements, the UK's short-to-medium investments in multidomain integration (MDI)/Joint All-Domain Operations (JADO)-type capabilities will need to be incremental, compared with the huge Joint All-Domain Command and Control (JADC2) programme underway in the US.

One potentially useful model for the UK is the US Navy's Naval Integrated Fire Control – Counter Air (NIFC-CA) programme, which has achieved an unprecedented level of real-time integration between surface warships, airborne assets and weapons in flight by focusing initially on connecting a limited number of core platforms essential to one mission set. To achieve similar results, the MoD will need to carefully prioritise early MDI investment tranches towards seamlessly connecting the assets that will gain the most combat effectiveness from the capability to share real-time situational awareness and conduct joint engagements with one another.

Selecting priority mission sets and using them as an analytical lens offers one way to identify the platforms and force elements where targeted investment in advanced cross-domain data sharing and third-party targeting capabilities would add the most tangible value in an operational context. The vignettes examined in this paper suggest that, in general, focusing on connecting key sensor assets such as F-35, E-7A and special operations forces teams, with long-range precision fires and integrated air and missile defence (IAMD) capabilities such as guided multiple launch rocket systems and Type 45, appears likely to significantly enhance combat effectiveness in challenging scenarios such as suppression of enemy air defence systems/destruction of enemy air defences (SEAD/DEAD) and IAMD, where the current force structure would struggle.

It is also worth noting that a necessary aspect of prioritisation is choosing which assets across the joint force do *not* currently need to be connected by an all-singing, all-dancing data sharing and automated command-and-control solution. Over-ambition coupled with a lack of focus in such a complex and interdependent effort involving all the armed services and Strategic Command is a recipe for programme failure.

Introduction

THE MINISTRY OF Defence (MoD) has made 'multidomain integration' (MDI) a conceptual cornerstone of UK military doctrine. In 2020 it released a Joint Concept Note (JCN 1/20) which set out an ambitious vision wherein the full capabilities of each service across all five domains would be able to function 'as a seamlessly integrated force [that] must also be fused across government and interoperable with principal allies'.[1] JCN 1/20 sets out multiple different areas where seamless integration will apparently be necessary across the joint force and wider government in order to be competitive in future conflicts. However, the same document admits that 'there is no fixed route to a known MDI destination, so this concept provides a headmark to allow us to explore and develop our MDI ambition'.[2] In other words, the UK currently has a conceptual aim, but no concrete plan for how to transform the force structure and processes that it currently has into ones that can undertake MDI.

A key contextual factor in the UK's MDI ambitions is a desire to retain conceptual and technical alignment with the US. Under the banner of Joint All-Domain Operations (JADO), the US military is attempting to transform the way in which it wages war, to better meet the threat posed by an increasingly powerful and assertive People's Republic of China. Each of the US armed services has ambitious programmes underway to revolutionise the way their assets exchange and use data, to heavily automate command-and-control (C2) processes, and to normalise real-time cooperative engagements across the joint force. The UK and other NATO allies are under pressure to follow the US lead at least to some degree, not least due to the need to retain interoperability with US forces as part of a warfighting coalition.[3] While the Pentagon recognises the political importance of retaining operational interoperability with allies, this objective takes second place behind the perceived imperative to radically improve American high-end capabilities against a rising Chinese threat in the Indo-Pacific. Given this, allies such as the UK need to understand the emerging American reforms in detail and plan investment to ensure that crucial integration points across the joint force remain technically and doctrinally compatible.

Chapter I of this paper demonstrates the scope of the US Department of Defense (DoD)'s ambitions for creating a fully fledged JADO capacity across the entire joint force, via its new Joint All-Domain Command and Control (JADC2) architecture. It briefly explains how the US Army's Project Convergence, the US Air Force's Advanced Battle Management System (ABMS)

1. Ministry of Defence (MoD), 'Joint Concept Note 1/20: Multi-Domain Integration', November 2020, p. iii, <https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/950789/20201112-JCN_1_20_MDI.PDF>, accessed 26 October 2022.
2. *Ibid*.
3. For more information, see Jack Watling, 'More Sensors Than Sense', in Justin Bronk and Jack Watling (eds), *Necessary Heresies: Challenging the Narratives Distorting Contemporary UK Defence* (Abingdon: Routledge, 2021), pp. 87–98.

and the US Navy's Naval Integrated Fire Control – Counter Air (NIFC-CA) and Project Overmatch programmes fit into the wider JADC2 effort.

Chapters II to IV use vignettes to examine which elements of the UK's future joint force might benefit most from JADC2-style connectivity across three specific mission sets. The conclusion explores the implied priorities and considerations for the MoD in pursuing its doctrinal aim of MDI through a capacity to conduct JADO.

It is important to note at the start that the scale and scope of the Chinese military threat in the Indo-Pacific, which is driving JADO ambitions in the US DoD, is of a different nature and magnitude to that faced by European NATO members. This has several important implications for what JADO will mean for the UK and its European NATO partner nations.

First, unless European NATO members wish to simply purchase JADC2-compliant off-the-shelf platforms, network architectures and weapons systems from the US, they will have to incorporate JADC2 compliance and compatibility into their own existing and future platforms and C2 systems. If they do not, over time they will lose operational interoperability with the US as the latter moves to ever more tightly integrated and automated data sharing, targeting and C2 arrangements to facilitate JADO.[4]

Second, both off-the-shelf procurement of JADC2-compliant equipment and modernisation of hardware, software and communications suites on existing platforms will necessarily take place at a much slower pace in Europe than in the US due to limited funding and other competing priorities. For the UK, regenerating the platforms required to deliver a warfighting division, purchasing sufficient munitions and funding adequate training and exercise activity is already overstressing the budget available – especially in light of the refusal of the government to commit to any increases in Defence spending beyond maintaining 2% of GDP in the Budget Autumn Statement.[5] As such, a comprehensive modernisation programme to make the whole UK Joint Force JADC2 compliant or otherwise capable of JADO-type activities is highly unlikely to be affordable for the foreseeable future.

Third, Defence in the UK (and in other European NATO member states) has limited programmatic capacity to staff and manage large capital programmes simultaneously. In the UK in autumn 2022 a major increase in the Defence budget was promised by then Prime Minister Liz Truss as a result of the invasion of Ukraine.[6] However, this increase was not confirmed in the Budget statement on 17 November. The Chancellor at the time acknowledged the need for increased

4. For more information, see Justin Bronk, *The Future of NATO Airpower: How Are Future Capability Plans Within the Alliance Diverging and How Can Interoperability Be Maintained?*, RUSI Whitehall Paper 94 (Abingdon: Taylor and Francis, 2019), pp. 67–73.
5. HM Treasury, 'The Autumn Statement 2022 Speech', 17 November 2022, <https://www.gov.uk/government/speeches/the-autumn-statement-2022-speech>, accessed 17 November 2022.
6. Malcolm Chalmers, 'From Famine to Feast? The Implications of 3% for the UK Defence Budget', *RUSI Occasional Papers* (September 2022), p. 1.

defence spending, but deferred a decision until the Integrated Review had been amended in light of the war in Ukraine.[7]

Prioritisation will, therefore, be critical to enable the UK to make efficient use of initial MDI investment tranches and to avoid decision paralysis in the face of the scale of the challenge. Chapters II, III and IV of this paper provide mission-centric vignettes to explore where the greatest potential combat effectiveness gains might be achieved by early investments to unlock JADO-style capabilities between specific UK platforms. The vignettes comprise: a campaign to suppress enemy air defence systems (SEAD) in Eastern Europe; the provision of IAMD to a forward-deployed UK force in the Middle East; and a contested freedom of navigation operation (FONOP). These were chosen as tasks that UK forces may be called upon to conduct, and also because they span the spectrum of conflict from high-intensity warfighting against a peer nation to potentially contested shows of presence during competition. The vignette approach was chosen to demonstrate that tying MDI/JADO questions to mission-focused output metrics can help policymakers move beyond necessary but broad and convoluted conceptual and doctrine discussions and towards identifying practical investment priorities. In addition to suggesting where significant value might be gained, even concise mission-focused vignettes can also shed light on where investment in cross-domain real-time data sharing is not likely to lead to transformative increases in effectiveness.

What are Joint All-Domain Operations?

JADO is a conceptual framework which has been created by the US DoD to describe its goal of integrated military operations across the domains of land, air, sea, space and the electromagnetic spectrum. The working US DoD definition of JADO is 'actions by the joint force in multiple domains integrated in planning and synchronised in execution, at speed and scale needed to gain advantage and accomplish the mission'.[8] The aim is to deliver integrated effects against an opponent across the phases of inter-state cooperation, competition and armed conflict.[9]

In other words, JADO is not a programme as such, but rather a guiding vision of the sort of operations the US armed forces are to be capable of undertaking in a future conflict. The intent is that by operating in this way, US forces will be able to create multiple simultaneous dilemmas for an enemy which cannot all be solved, and which compel hostile troops and commanders to make difficult or impossible trade-offs. A critical component of this vision is the simplification and acceleration of current decision-making processes, moving from planning cycles measured in days to hours or even minutes. In some respects, this represents a simple desire to speed up the Observe Orient Decide Act loops which govern military decision-making at all levels.

7. HM Treasury, 'The Autumn Statement 2022 Speech'.
8. US Air Force and US Space Force, 'The Department of the Air Force Role in Joint All-Domain Operations', Air/Space Force Doctrine Publication 3-99, 19 November 2021, p. 4, <https://www.doctrine.af.mil/Doctrine-Publications/AFDP-3-99-DAF-Role-in-Jt-All-Domain-Ops-JADO/>, accessed 16 March 2022.
9. *Ibid*., p. 3.

This has led some to suggest that there is little to differentiate JADO from earlier operational concepts.[10] As a top-down set of guiding principles, JADO is also very similar to the vision laid out by the UK's own MDI concept note.[11]

However, the US has a practical strategy which the DoD is pursuing to enable – called JADC2. The working definition of JADC2 in the DoD is 'the warfighting capability to sense, make sense, and act at all levels and phases of war, across all functional areas, domains, and with partners, to deliver information advantage at the speed of relevance'.[12] In more simple terms, JADC2 is a strategy which aims to connect all the sensors across the US Air Force, Navy, Army and Marine Corps into a single and highly automated network.[13] Its primary function is to provide a new kind of joint decision-making and disseminating tool for commanders to enable them to plan, orchestrate and execute JADO.[14] This vision of JADC2 is, to put it mildly, an extremely complex and ambitious undertaking. However, a unified decision-making tool is not the limit of the ambitions for JADC2 across the DoD.

For many, JADC2 is also a way to develop a highly automated 'any-sensor, any-shooter' network that will process and disseminate data between all the platforms within a joint force, and automatically allocate fire missions and other non-kinetic tasks to the optimal unit or platform at any given time. In this vision, the key output of JADC2 would be a version of a combat cloud or an Internet of Things, which has been described as an application enabled by modern digital communications networks to interface with connected sensors and shooters in a similar fashion to the Uber ride-hailing app.[15] In this vision, a 'shooter' does not know or need to know where the target track that they have been automatically allocated was gathered from, and nor would the original source of that target data need to know which 'shooters' were available to prosecute the target when they sent the request for fires. Instead, the system would optimise the matching of detected target tracks and weapons solutions automatically.

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10. Ewan Sime, 'How All Domain Is JADO?', 360iSR, 16 March 2021, <https://www.360isr.com/how-all-domain-is-jado>, accessed 21 February 2022.
11. MoD, 'Joint Concept Note 1/20'.
12. US Department of Defense, 'Summary of the Joint All-Domain Command and Control (JACD) Strategy', March 2022, p. 2, https://media.defense.gov/2022/Mar/17/2002958406/-1/-1/1/SUMMARY-OF-THE-JOINT-ALL-DOMAIN-COMMAND-AND-CONTROL-STRATEGY.PDF>, accessed 26 October 2022.
13. John R Hoehn, 'Joint All-Domain Command and Control (JADC2)', Congressional Research Service, updated 21 January 2022, p. 1, <https://sgp.fas.org/crs/natsec/IF11493.pdf>, accessed 26 October 2022.
14. Bryan Clark and Dan Patt, 'JADC2 May be Built to Fight the Wrong War', *Breaking Defense*, 14 January 2021, <https://breakingdefense.com/2021/01/jadc2-may-be-built-to-fight-the-wrong-war/>, accessed 16 March 2022.
15. Theresa Hitchens, 'All-Domain Ops Require Rethinking Combatant Commands: Goldfein', *Breaking Defense,* 10 March 2020, <https://breakingdefense.com/2020/03/all-domain-ops-require-rethinking-combatant-commands/>, accessed 10 March 2022.

This would represent a major shift from network-centric warfare to data-centric warfare. It requires a C2 system that can be platform-agnostic and capable of taking data from any sensor and supplying firing and targeting solutions to any available effector systems.[16] It is critical to understand what this entails; most networked systems are designed to receive data from a specific sensor. For example, an air defence system is designed to receive target acquisition and tracking data from its own bespoke radars, whose data is formatted for that system's fire control and battle management systems. The concept of JADC2 holds that the overarching battle management system and C2 network supporting multidomain operations should be able to rapidly receive, interpret, weight and assign actions in response to any data it receives and distribute it in a format that is usable by any other asset or weapons system connected to the network.

16. Jerome Dunn, Director Strategic Missile Solutions at Northrop Grumman, speaking at the 2022 RUSI Missile Defence Conference, London, 23 February 2022.

I. Where is JADO Now?

THE CONCEPT OF JADO is being explored through a series of experiments led by the US services that are designed to examine how it can be realised under the aegis of JADC2. JADC2 is both an approach and a procurement effort; it requires the US services to establish common data standards so that information can be shared seamlessly between them, as well as the procurement of new communications architectures that allow that information to be shared. However, it also requires a change in C2 culture with greater thought given to the synchronised, optimal allocation of resources, force positioning and decision-making. There are already several technical efforts underway within each service, which sit underneath the overall JADC2 effort.

The US Air Force is essentially the lead service on JADC2, with its primary efforts in this regard being channelled into its ABMS programme. ABMS is being developed in line with the Joint Warfighting Concept that will shape the doctrines created by the US services to define their role in JADO.[17] JADC2 is the system that will enable US commanders to connect any sensor system to any effector. It is notionally a joint effort, although US Air Force officials have stressed that the technologies developed by the Army and Navy are designed to dovetail into the ABMS.[18] In addition to integrating C2 efforts from across the US armed forces, JADC2 is also expected to integrate the US nuclear arsenal into the same network of sensors and effectors, although details on that effort are classified.[19]

ABMS itself was originally intended to replace the E-3 AWACS before it was expanded to become a programme that will procure equipment and software, as well as implement new C2 and data policies.[20] The US Air Force's experiments within ABMS have demonstrated the ability to transmit data from Army and Navy radars to F-22 Raptor and F-35 Lightning fighter aircraft; it has been used in defence of US space assets and to engage cruise missiles.[21] The US Air Force has also demonstrated the use of a KC-46 Pegasus tanker aircraft as a communications node that was used to relay information from fourth- to fifth-generation aircraft.[22] The KC-46

17. Hitchens, 'All-Domain Ops Require Rethinking Combatant Commands'.
18. Carlo Munoz, 'Clear Channel: The Evolution of the Pentagon's Joint All-Domain Command and Control Scheme', *Janes International Defence Review,* April 2022.
19. Colin Demarest, 'How JADC2 is Improving Nuclear Command and Control', *Defense News,* 9 March 2022, <https://www.defensenews.com/battlefield-tech/it-networks/2022/03/09/how-jadc2-is-improving-nuclear-command-and-control/>, accessed 3 May 2022.
20. John R Hoehn, 'Advanced Battle Management System (ABMS)', Congressional Research Service, updated 15 February 2022, p. 1, <https://sgp.fas.org/crs/weapons/IF11866.pdf>, accessed 3 May 2022.
21. *Ibid*.
22. *Ibid*.

application of the ABMS programme has become the first material step, taking it from a purely experimental standing to an in-service capability.[23]

The US Army has launched Project Convergence, which is seeking to test the ability of national-level intelligence assets to coordinate and direct long-range fires with the assistance of AI.[24] Project Convergence was established by the US Army Futures Command to test a single use case; a sensor-to-shooter link to enhance and test the ability of in-service systems to pass data between themselves.[25] Its first iteration in 2020 proved that sensor-to-shooter time could be reduced from an average of 14 minutes to a few seconds. However, it also revealed that many of the systems in service with the US armed forces could not pass data to each other in compatible forms, leading to the realisation that the network that facilitates the flow of information is the centre of gravity as far as JADO is concerned.[26] Project Convergence has since expanded to seven use cases and the establishment of the Joint Systems Integration Lab, which is charged with practising the movement of data between systems. Its work has been deemed vital to realising JADC2.[27]

The US Army's technical contribution in the land domain is the Integrated Tactical Network (ITN), which has experimented with 4G and 5G and hybrid cloud architectures to support mounted and dismounted operations, including the movement of large data sets with low latency.[28] The US Army has also moved its integrated air and missile defence (IADS) Battle Command System (IBCS) into low-rate initial production. The system enables assets from multiple domains to be integrated into a single air defence network. When provided with a suitable relay and datalink translation node (in this case an 'EMC2' network bridge on a U-2),[29] IBCS has been used to coordinate an engagement between F-35 and the Army Field Artillery Tactical Data System.[30] In

23. Charles Pope, 'With its Promise and Performance Confirmed, ABMS Moves to a New Phase', *Air Force*, 21 May 2021, <https://www.af.mil/News/Article-Display/Article/2627008/with-its-promise-and-performance-confirmed-abms-moves-to-a-new-phase/>, accessed 3 May 2022.
24. Ashley Roque, 'Project Convergence 2020: US Army Hosting Kill Chain Demonstration', Janes, 16 September 2020.
25. Lieutenant General James Richardson, speaking at Project Convergence: An Experiment for Multidomain Operations, event hosted by Center for Strategic and International Studies (CSIS), 16 February 2022, <https://www.csis.org/events/project-convergence-experiment-multidomain-operations>, accessed 8 March 2022.
26. *Ibid*.
27. Sydney J Freedberg Jr, 'New Army Lab Tests Joint Tech: Project Convergence', *Breaking Defense*, 20 April 2021, <https://breakingdefense.com/2021/04/new-army-lab-tests-joint-tech-project-convergence/>, accessed 8 March 2022.
28. Carlo Munoz, 'Army ITN Experiments to Showcase JADC2 Technologies', Janes, 29 January 2021.
29. Lockheed Martin, 'Lockheed Martin Flies Real-Time, Mission Enabling Kubernetes Onboard U-2', 14 December 2020, <https://news.lockheedmartin.com/lockheed-martin-flies-real-time-mission-enabling-kubernetes-onboard-u-2>, accessed 26 October 2022.
30. Giles Ebbutt, 'Connecting Air Defences: US Army Prepares to Bring IBCS Online,' *Janes International Defence Review,* April 2022.

the land domain at least, experimentation with JADO has made clear that passing data between systems in a usable form is the primary challenge to be overcome and emphasises that this is the centre of gravity around which the success of the project will turn.

The US Navy is focused on Project Overmatch, a secretive effort about which little is known except for the intention to introduce an AI-enabled network architecture and facilitate distributed operations through manned–unmanned teaming.[31] Project Overmatch builds on the progress already made under the NIFC-CA programme, which was developed as an advanced version of what was initially termed 'cooperative engagement capability' (CEC) whereby aircraft and surface ships could share target track information and guide each other's weapons in flight.[32] In its fully operational state the NIFC-CA architecture currently connects surface combatant ships equipped with Aegis Baseline 9 or a later version of the powerful Aegis air and missile defence system to the F/A-18E/F Super Hornet and EA-18G Growler fast jets, and E-2D Advanced Hawkeye AWACS aircraft of the carrier air group (CAG).[33] Its primary function is to create a fused recognised air picture which is fed by all the radars on ships and in the air back to the operations centre on the aircraft carrier itself. This allows a huge increase in the effective resolution and coverage of the whole fleet's radar coverage. It is especially effective at increasing detection ranges against small or stealthy targets such as advanced anti-ship missiles and fifth-generation fighter aircraft due to the ability to exploit the dispersed positions of the various contributing radar sensors for multi-static triangulation. As a secondary function, NIFC-CA was designed to enable assets in the task group to designate targets for and guide each other's missile systems.[34]

The way in which the US Navy developed NIFC-CA is potentially highly instructive for the UK and other Allied nations looking to invest in similar capabilities in their own forces. Rather than making NIFC-CA into a major acquisition programme of record, the US Navy inserted common network connectivity and data standards into the requirements within the major acquisition/modernisation programmes for specific platforms that were judged most critical for a specific

31. Munoz, 'Clear Channel'.
32. Nicholas A O'Donoghue, Samantha McBirney and Brian Persons, *Distributed Kill Chains: Drawing Insights for Mosaic Warfare from the Immune System and from the Navy* (Santa Monica, CA: RAND Corporation, 2021), p. 43; Theresa Hitchins, 'Services' Fear Of Losing "Control" Blocking JADC2 Progress, Experts Say', *Breaking Defense*, 22 October 2020, <https://breakingdefense.com/2020/10/services-fear-of-losing-control-blocking-jadc2-progress-say-experts/>, accessed 30 May 2022.
33. Navy Recognition, 'Raytheon's SM-6 Surface-to-Air Missile Once Again Shatters Engagement Distance Record', October 2016, <https://www.navyrecognition.com/index.php/newsb/defence-news/2016/october-2016-navy-naval-forces-defense-industry-technology-maritime-security-global-news/4406-raytheons-sm-6-surface-to-air-missile-once-again-shatters-engagement-distance-record.html>, accessed 30 May 2022.
34. Program Executive Office Integrated Warfare Systems, 'Navy Conducts First Live Fire NIFC-CA Test with F-35', *Naval Sea Systems Command News*, 13 September 2016, <https://www.navsea.navy.mil/Media/News/Article/942188/navy-conducts-first-live-fire-nifc-ca-test-with-f-35/>, accessed 30 May 2022.

mission – in this case counter-air and missile defence. The mission hinged most critically on the forward sensors on the E-2C/D Hawkeye AWACS, the later cancelled Joint Land Attack Cruise Missile Defense Elevated Netted Sensor System (JLENS) and F/A-18E/F Super Hornet, and the SM-3 and SM-6 'shooters' used by the Aegis combat system onboard US guided missile destroyers and cruisers. Therefore, these systems were prioritised in terms of investment in unlocking shared situational awareness and third-party targeting capabilities.[35]

Since NIFC-CA was not a major acquisition platform for a poorly defined future connectivity goal, but rather a budgetarily small programme with the power to insert requirements into pre-existing equipment acquisition or modernisation programmes, it did not fall foul of any of a series of major budget cuts over the decade of primary development.[36] Even when JLENS was cancelled, this did not endanger NIFC-CA as a whole. Furthermore, once the cooperative engagement capabilities and combined situational awareness enabled by NIFC-CA in its early form had been demonstrated, it was easier to both justify the funding for and articulate the requirements needed to integrate other platforms, such as F-35C. Consequently, NIFC-CA is currently delivering real-time shared situational awareness and cooperative engagement capabilities between US Navy vessels, aircraft and weapons far in excess of what has yet been achieved by more ambitious but less well-defined US Air Force programmes, such as ABMS. The more ambitious Project Overmatch builds on the progress already made by the US Navy with an incremental and mission-specific approach to unlocking increased combat effectiveness through cross-domain connectivity.

Compared with this sort of operational capability, tangible UK progress in making JADO-style capabilities a practical reality remains largely experimental. The MoD has specified a guiding ambition to develop the capability to conduct MDI in doctrine and strategy publications such as JCN 1/20, and the Integrated Operating Concept.[37] However, most practical investment is currently being undertaken by experimental establishments within each service. One example is the RAF's Rapid Capabilities Office (RCO), which has taken a somewhat different approach from the US to the problem of joint force connectivity.

Instead of looking to build connectivity through feeds and common applications to link platforms or services, the RCO has focused initially on creating a common data environment called Nexus. The idea behind Nexus is to provide a virtual environment which acts as a Defence-owned portal where other platforms, networks, sensors and applications can both request and feed in data, according to published integration standards. Nexus concentrates on metadata so that applications and platforms alike can see what and who is doing what and where (at varying classifications) across the network, without having to off-board data wholesale as a

35. O'Donoghue, McBirney and Persons, *Distributed Kill Chains,* p. 43.
36. *Ibid*., p. xix.
37. MoD, 'Joint Concept Note 1/20'; MoD, 'Integrated Operating Concept', 2021, <https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1014659/Integrated_Operating_Concept_2025.pdf>, accessed 26 October 2022.

standard procedure.[38] This keeps baseline bandwidth demands low, and ensures that when a connected user specifically requests detailed data, there is sufficient capacity to pass it through the network along paths which have already been defined using the basic metadata. The RCO conducted flight trials with a hand-carried Nexus portal and a virtual communications hub application called Raven in 2021 on a Voyager tanker, and demonstrated the ability to pull data from satellite communications and a range of other sources into a common operating picture in real time.[39]

While the creation of such a virtual data indexing and exchange medium is a significant step, it remains to be seen if it can be scaled up to meet the requirements of the whole force. It also does not by itself solve the problem of how to generate sufficiently low-latency real-time connectivity between joint force assets to enable third-party weapons cueing or guidance. Therefore, Nexus does not fill the same niche for the UK as NIFC-CA does for the US Navy or ABMS does for the US Air Force.

38. Author interview with Air Commodore Jez Holmes and Air Information Experimentation Laboratory experts, Rapid Capabilities Office, Farnborough, 21 June 2022.
39. Royal Air Force, 'RAF Rapid Capabilities Office Demonstrates New Technologies Developed With Industry Partners', 12 March 2021, <https://www.raf.mod.uk/news/articles/raf-rapid-capabilities-office-demonstrates-new-technologies-developed-with-industry-partners/>, accessed 22 June 2022.

II. Vignette 1: SEAD in Eastern Europe

SINCE THE RUSSIAN invasion of Ukraine began on 24 February 2022, the once-theoretical spectre of a confrontation with Russian conventional forces in Eastern Europe has gone from the highest-risk/lowest-likelihood scenario at the top of NATO threat estimates to an unavoidable priority for future force planning.

The performance of the Russian aerospace forces' fixed-wing and rotary assets during the initial two months of the war was notably poor – Russia failed to destroy the Ukrainian air force, or exert significant influence on the battles on the ground.[40] This was, in large part, due to its inability to conduct effective suppression/destruction of enemy air defences (SEAD/DEAD). Ukraine's continued ability to operate its various mobile surface-to-air missile (SAM) systems forced Russian aircraft to operate at very low altitude to take advantage of radar-horizon limitations and terrain for cover, but this left them vulnerable to anti-aircraft artillery and man-portable air defence systems (MANPADS) fire.[41] The majority of combat air losses on both sides during the first phase of the war have been caused by SAMs and MANPADS – with the latter posing a threat precisely because the presence of the former forces both sides to operate at very low level.[42]

This should prompt renewed prioritisation of SEAD/DEAD capability for NATO air forces, since at present only the US Air Force can field credible answers to the sorts of mobile SAM tactics used by both sides in Ukraine.[43] Therefore, the first vignette in this paper examines where targeted investment in JADO-style capabilities within the UK's joint force might add greatest benefit during a SEAD/DEAD operation against Russian forces in a geographically constrained flashpoint confrontation in Eastern Europe.

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40. Justin Bronk, 'Is the Russian Air Force Actually Incapable of Complex Air Operations?', *RUSI Defence Systems*, 4 March 2022,< https://www.rusi.org/explore-our-research/publications/rusi-defence-systems/russian-air-force-actually-incapable-complex-air-operations>, accessed 11 April 2022.
41. Justin Bronk, Nick Reynolds and Jack Watling, 'The Russian Air War and Ukrainian Requirements for Air Defence', RUSI Special Resource, 7 November 2022, Chapter I.
42. *Ibid*.
43. Justin Bronk, 'Getting Serious About SEAD: European Air Forces Must Learn from the Failure of the Russian Air Force over Ukraine', *RUSI Defence Systems*, 6 April 2022, <https://www.rusi.org/explore-our-research/publications/rusi-defence-systems/getting-serious-about-sead-european-air-forces-must-learn-failure-russian-air-force-over-ukraine>, accessed 11 April 2022.

The UK and its allies would face three primary operational challenges during any SEAD/DEAD operation. The first would be detecting, identifying, mapping and then targeting the key long-range sensors, SAM systems and command nodes which provide the overall coordination for an IADS and allow it to threaten assets at long ranges.[44] The second would be detecting, suppressing and then destroying the more numerous medium- and short-range mobile SAM batteries and systems which move with enemy ground forces.[45] The third challenge would be achieving sufficient effect against these two categories of targets (and their crews) to enable air superiority to be achieved, without taking unacceptable attrition in airframes or running out of critical munitions in the process. Any enemy air superiority assets would also need to be tracked and engaged if they attempted to engage UK or allied SEAD/DEAD assets, although NATO proficiency in the offensive/defensive counter-air mission set is already sufficient to provide a level of overmatch against any likely aerial adversary.

Time pressure is also likely to be a major factor, since the UK and most other Western nations depend heavily on the firepower and ISR delivered by air forces to allow their relatively small ground and maritime forces to achieve success against enemies able to field greater mass and firepower.[46] Therefore, there would be heavy time pressure to achieve SEAD/DEAD results so that close air support and airborne ISR aircraft could be used freely within the area of operations. To work out where the greatest benefits might be gained from investment to allow joint, cooperative real-time engagements in such an operation, it is first necessary to identify the critical sensor and shooter platforms across the joint force.

Fortunately, the potentially critical assets within the UK's joint force are relatively easy to identify for any SEAD/DEAD campaign. For the first challenge of detecting, identifying and mapping the key long-range sensors and C2 nodes, standoff signals intelligence (SIGINT), electronic intelligence (ELINT) and radar-surveillance platforms – most obviously the RC-135W Airseeker and the new E-7A Wedgetail – would be critical in the run-up to any operation. However, once combat starts these aircraft would need to operate from beyond the reach of the long-range S-400 or S-300V4 SAM systems within a Russian IADS. This is because of their large signature, need to operate at high altitudes where long-range SAMs are most effective, and limited self-protection or evasion capabilities. Furthermore, many SIGINT and ELINT intercepts require significant processing and analysis before they can be fully exploited to help map an IADS and so incorporating an asset such as RC-135W into a JADC2-style real-time data sharing and targeting architecture might add only relatively marginal value compared to current processes. E-7A Wedgetail, by contrast, will be taking over from the E-3D Sentry as a key airborne battlespace management and C2 node within whatever new architectures are designed for combat air and joint forces use. As such, despite its likely position several hundred kilometres from the front lines during combat operations, it will remain a key communications relay and C2 node during active SEAD/DEAD

44. For more detail, see Jack Watling, Justin Bronk and Sidharth Kaushal, 'A UK Joint Methodology for Assuring Theatre Access', *Whitehall Report,* 4-22 (May 2022), pp. 7–22.
45. *Ibid*., pp. 30–44.
46. For more information, see Jack Watling, 'The Future of Fires: Maximising the UK's Tactical and Operational Firepower', *RUSI Occasional Papers* (November 2019).

operations, especially as a connective node between the air component and other elements of the force. Therefore, the E-7A is an example of where investing in radical improvements in the ability to exchange data in real time with forward sensors and long-range fires assets across all domains is likely to result in significant efficiency gains for the whole force.

Once combat operations begin, the most critical sensor nodes will be those that can detect, identify and track high-value enemy sensors, SAMs and C2 nodes from inside the threat range they pose. Obvious candidates within the UK's current order of battle include the F-35B Lightning II and infiltrated special operations forces (SOF) teams. Both have the potential capability to get relatively close to threat systems covertly and can carry sophisticated sensors. The F-35's sensor suite is particularly well suited to both hunting key nodes within an IADS and enabling more routine SEAD/DEAD efforts against mobile SAM systems in an ongoing campaign. This is because its sensors and software allow even a single aircraft to rapidly triangulate the position of a hostile SAM radar in flight more effectively than a whole flight of previous-generation 'Wild Weasel' F-16CJs with dedicated SEAD sensors.[47]

Both the F-35 and SOF teams have limited capabilities for direct kinetic attacks due to their relatively small numbers within the overall UK force structure, the need to husband limited munitions carriage capacity for key targets, and the need to avoid detection while deep in hostile territory. In other words, there is a significant mismatch between the ISTAR capabilities and the level of lethality which these assets can project into a hostile IADS when working alone.[48] Therefore, in a SEAD/DEAD context, both F-35B and infiltrated SOF teams are likely to represent key UK sources of real-time situational awareness about key IADS targets, and also force elements whose lethality and survivability would be greatly increased by the ability to call in long-range firepower from other assets in real time.

To assess which potential 'shooters' would bring the most benefit from JADC2-style connectivity with these forward sensors in a SEAD/DEAD operation, the first question is which platforms are likely to be able to deploy standoff weapons from sufficient range? Shooters must be able to threaten targets without unacceptable risk to sustainably leverage the situational awareness generated by the F-35Bs, SOF teams or other forward sensors for increased lethality. Within the UK joint force there are limited potential options, which makes prioritisation for investments in JADO-style cooperative engagement capabilities easier.

One clear candidate would be the guided multiple launch rocket systems (GMLRS) which currently provide the British Army's long-range artillery firepower. GMLRS can fire rockets out to 70 km with submunition warheads that make each one lethal against lightly armoured vehicles across a spread pattern on impact.[49] If the British Army's GMLRS were equipped with such munitions,

47. Author briefings with F-35 test pilots, Fort Worth, Texas, 19 February 2019.
48. Confirmed in face-to-face author interviews with RAF and Royal Navy F-35 pilots, RAF Marham, 20 September 2022.
49. Army Technology, 'MLRS (Multiple Launch Rocket System)', 10 September 2021, <https://www.army-technology.com/projects/mlrs/>, accessed 3 May 2022.

they would potentially be ideal for use against SAM batteries in the field. Furthermore, the UK has signalled its intention to procure the precision strike missile (PrSM) for its GMLRS launchers, which would provide a long-range precision strike capability of 499 km.[50] PrSM has also been tested with a multi-mode seeker that would enable the missile to track and engage radars, which is expected to be included in the Spiral 1 upgrade of the missiles.[51]

Other options could include Royal Navy surface or submerged vessels firing Tomahawk land attack missiles (TLAMs), or RAF Typhoons with Storm Shadow cruise missiles. However, due to their high cost and consequently limited inventory capacity, large cruise missiles such as Storm Shadow and TLAMs are likely to be reserved for high-value targets during a SEAD/DEAD mission such as long-range target acquisition radars, command centres and command vehicles. Therefore, they might benefit from real-time target data from penetrating ISTAR assets during the initial phases of any campaign, but would be less critical during the later DEAD phases, where the primary targets are likely to be mobile medium- and short-range SAM batteries. Against these targets, less costly and shorter-ranged munitions would be needed for viable mass of fires to be achieved. Therefore, likely candidates would be RAF Typhoons with SPEAR 3 (if it were purchased for the Typhoon Force), conventional Army AS90 or other 155-mm artillery batteries within range of SAM systems near the enemy front lines, and possibly AH-64E Apache gunships equipped with Hellfire or JAGM missiles. The point is that potential priorities for investment if SEAD/DEAD is chosen as a policy-relevant mission can be identified relatively easily by assessing where lethality would be most significantly increased through tactical real-time sensor-shooter integration across domains, compared to more traditional joint operations integrated at the operational or campaign level.

Aside from increased lethality, another aspect to potentially consider is which platforms would benefit most in terms of survivability from JADC2-style connectivity during a SEAD/DEAD operation. This is more difficult to define than lethality enhancement, since the survivability of any given platform is based more on contextual factors, including the risk appetite of both the attacking and defending force and the tactics each employ. However, a basic starting point would be to note that in a SEAD/DEAD operation the assets that might see increased survivability from JADC2-style connectivity are those which will be expected to operate within the missile engagement zone (MEZ) of enemy SAM systems (and possibly aircraft).

During the competition phase, before kinetic warfare begins, ISTAR assets conducting mapping of the IADS and airspace monitoring may well be operating within the MEZ of long-range enemy SAMs. The same may be true of tanker aircraft conducting aerial refuelling for both ISTAR and combat air patrol orbits close to adversary territory. If they perceive a direct conflict as

50. Albert L, 'British Army Confirms Intentions for Precision Strike Missile Procurement', *Overt Defense,* 2 February 2021, <https://www.overtdefense.com/2022/02/02/british-army-confirms-intentions-for-precision-strike-missile-procurement/>, accessed 4 May 2022.
51. Timothy Bucklin, 'US Army Tests Multi-Mode Seeker for PrSM Used to Hunt Enemy Radars', *Overt Defense,* 9 June 2020, <https://www.overtdefense.com/2020/06/09/us-army-tests-multi-mode-seeker-for-prsm-used-to-hunt-enemy-radars/>, accessed 4 May 2022.

inevitable or desirable, the adversary might attempt to open hostilities by conducting long-range SAM launches at these high-value enablers while they are within range. Given the lack of agility and limited self-defence options available to airliner-derived platforms, the ability to receive the earliest possible warning of hostile missile launches from any available sensor is critical to increasing their odds of survival. The Voyager MRTT in particular will be a key part of any air operation due to the critical nature of aerial refuelling capabilities for all fast jet and most ISTAR operations. Once shooting starts, tanker orbits will be pushed significantly further back from potential threats, which will keep them safer but means that each fuel offload grants less time on-mission for combat air assets. The centrality of the Voyager MRTT to almost all significant RAF air operations, and its consequent ubiquitous presence in the battlespace, is why it has already been chosen for airborne trials of the new Nexus common data platform and Raven communications node.[52]

To enhance the survivability of key assets such as the Voyager MRTT, the most valuable airborne connection in a SEAD context is likely to be the F-35, since its electro-optical/infra-red surveillance suite can detect, classify and track missile and rocket launchers from hundreds of kilometres away in any direction around the aircraft.[53] Ground-based long-range missile defence radars such as the MPQ-65 and AN/TPY-2 used by allied Patriot and THAAD batteries could also provide advance warning to airborne assets of long-range SAM launches due to their high flight apexes and consequent similarity to ballistic missiles. As will be seen in Vignette 2, these missile defence assets are also likely to benefit in situational awareness terms from being linked via such enablers to forward air assets for their own mission sets. Orbital missile launch detection and tracking capabilities would also be a potentially useful source of launch warning information if they were able to process the signatures of long-range SAMs and relay warnings in real time.

This vignette is not intended to imply that SEAD/DEAD is necessarily one of the critical missions against which initial UK investments in enabling practical JADO-style capability should be judged. Rather, it is intended to illustrate that considering how the existing and projected force structure would need to be employed against a notional mission set can guide analysis towards the assets that would likely give the greatest return on investment in improved connectivity across domains. In the case of the SEAD/DEAD mission, even a superficial examination of the force structure suggests that necessarily limited early investment could deliver significant increases in combat effectiveness without needing to achieve whole force transformation along the lines of American JADO efforts. The UK has a small number of assets that could be tasked with covertly penetrating defended areas, and which can carry potent sensors, but have limited organic firepower. It also has a limited range of assets that can deploy long-range firepower capabilities but which are reliant on off-board targeting information to hit elusive targets from a safe distance in a high-intensity scenario. Therefore, connecting these two sets of capabilities

52. Jon Lake, 'Operating at the Speed of Relevance', *Armada International*, 9 May 2022, <https://www.armadainternationalcom/2022/05/operating-at-the-speed-of-relevance/>, accessed 22 June 2022.
53. Kris Osborne, 'Enemies Beware: America's F-35 Might Soon Track Ballistic Missiles', *National Interest*, 2 April 2021.

would be a natural place to invest in delivering an initial real-time cross-domain connectivity programme for the UK if SEAD/DEAD is assessed as an important mission set. The survivability of critical enabler nodes, such as tankers, which lack organic self-defence and evasion capabilities, could also potentially be significantly enhanced by being connected to force elements able to provide early warning of enemy long-range missile launches. They would likely also be present in theatre anyway due to their key roles. Therefore, they might also make logical sense as a priority for early upgrades to enhance their real-time connectivity with other force elements.

In summary, a brief examination of the critical assets for a notional SEAD/DEAD campaign suggests that if this mission were chosen as a key planning task, the MoD could usefully prioritise initial MDI/JADO-style cross-domain connectivity investment towards penetrating assets, long-range precision fires capabilities and large enabler aircraft.

III. Vignette 2: Integrated Air and Missile Defence

THIS VIGNETTE EXAMINES a notional scenario where the UK is required to provide an element of an IAMD capability for a European force that has been forward deployed to the Middle East in an uncertain political climate with an evolving insurgency led by a regional actor. Across current and future deployments of this type, the threat to British and allied troops deployed on the ground is likely to be increasingly significant and sophisticated. In any political environment where tolerance of casualty risk is limited, having a credible IAMD solution for forward deployed forces is likely to be increasingly essential for operational viability. However, providing IAMD to forward deployed forces is a complex challenge.

The threat environment has developed considerably over the past decade due to the democratisation of small unmanned aerial systems (sUAS) and the proliferation of larger remotely piloted aerial systems (RPAS) that can be used as munitions in their own right and also to rapidly cue in long-range fires with precision. Whereas past forces would have to contend only with an opponent's air force and incoming rocket artillery, a European force deployed anywhere in the world today will likely face threats from armed sUAS and RPAS, as were encountered by Russian forces deployed to Khmeimim Airbase in 2018.[54]

More complex missile threats have also proliferated. Houthi attacks against critical national infrastructure in Saudi Arabia have often come from different directions simultaneously, and been conducted across a wide geographical area, forcing the available Saudi air defences to be dispersed, thus diminishing their efficacy.[55] Iran itself has demonstrated the ability to conduct attacks against American bases; the 2020 ballistic missile strikes against the Al Asad air base in western Iraq are the most prominent demonstrations of this capability. Sixteen ballistic missiles were fired at the base in retaliation for the assassination of Iranian General Qassem Soleimani, 11 of them – each carrying a 1,000-lb warhead – hit the base, causing extensive damage.[56]

Consequently, the future threat environment for UK forces deployed to the region is likely to be characterised by a mix of massed sUAS and loitering munitions, as well as higher-end cruise and ballistic missiles, used together in such a way as to increase the complexity of the IAMD challenge. Given this, more traditional siloed deployments of air and missile defence systems, such as Phalanx-B to guard Camp Victory in Iraq from rockets, artillery and mortar rounds, or

54. Anton Mardasov, 'How Russia Defends its Syrian Air Base from Attacks', *Al-Monitor,* 31 May 2019, <https://www.al-monitor.com/originals/2019/05/russia-khmeimim-syria-turkey-hts-drones>, accessed 12 May 2022.
55. Tyler Rogoway, 'The Strike on Saudi Oil Facilities was Unprecedented and it Underscores Far Greater Issues', *War Zone*, 1 December 2019.
56. David Martin and Mary Walsh, 'Who Would Live and Who Would Die: The Inside Story of the Iranian Attack on Al Asad Airbase', *CBS News,* 8 August 2021.

Patriot PAC-3 batteries to reduce the threat to Eastern European allies from short-range ballistic missiles in 2022, are likely to be insufficient.[57] No one system can efficiently or effectively cover the threat spectrum adequately. It follows that enhancing the ability of UK systems to connect to a common recognised air picture, and the ability to link together and cross-cue different sensors and effectors for layered effects, is likely to offer the best means to increase combat effectiveness for IAMD coverage.

The primary challenge in any IAMD scenario remains the detection and tracking of incoming threats, and so protecting a UK deployment in the Middle East against the sorts of potential threats outlined would require the provision of persistent omni-directional air situational awareness across all the likely incoming threat altitudes, ranges and speed bands. This is an extremely difficult requirement to meet within the UK's current order of battle, but in principle it requires the ability to combine the outputs from as many of the available ground-, maritime- and air-based radars and other sensors as possible into a common situational awareness picture. Obvious candidates within the existing force structure include the Giraffe Agile Multi-Beam radar that forms a core part of the British Army's Sky Sabre medium-range air defence system, the Multi-role Electronically Scanned Array (MESA) radar mounted on the RAF's new E-7A Wedgetail AWACS aircraft, the Sampson radar on any Type 45 destroyers operating in littoral waters nearby, and the smaller radars carried by counter-unmanned aerial systems (C-UAS) solutions such as Drone Dome or ORCUS.[58]

The fact that the IAMD threat encompasses such a broad range of target sizes, speeds and ranges means that operational effectiveness is likely to be disproportionately improved if systems optimised to cover different threat bands and altitudes can be linked together to create a combined situational awareness picture in real time. For threats coming in at close to ground level, sUAS and loitering munitions are likely to be a particularly challenging problem set, since these systems often have a low radar cross section and very small electro-optical/infra-red signature, and can travel extremely slowly. These properties, together with clutter from uneven ground, vegetation and buildings, make them particularly difficult to track and engage with

57. George Allison, 'C-RAM Firing in Response to a Rocket Attack Launched by Insurgents in Basra in 2009', *UK Defence Journal*, 4 February 2016, <https://ukdefencejournal.org.uk/3604-2/>, accessed 18 May 2022; Gastón Dubois, 'Germany Reinforces NATO's Eastern Flank by Deploying Patriot Systems in Slovakia', *AviacOnline*, 17 March 2022, <https://www.aviacionline.com/2022/03/germany-reinforces-natos-eastern-flank-by-deploying-patriot-systems-in-slovakia/>, accessed 26 October 2022.

58. SAAB, 'Giraffe-AMB', <https://www.saab.com/products/giraffe-amb>, accessed 26 October 2022; Northrop Grumman, 'Multi-Role Electronically Scanned Array (MESA) Surveillance Radar', <https://www.northropgrumman.com/what-we-do/air/multi-role-electronically-scanned-array-mesa-surveillance-radar/>, accessed 26 October 2022; BAE Systems, 'Sampson Multi-Function Radar', <https://www.baesystems.com/en-uk/product/sampson-multi-function-radar>, accessed 26 October 2022; MoD and Defence Equipment and Support, 'Counter-Drone Tech and State-of-the-Art Radar for the RAF', <https://www.gov.uk/government/news/counter-drone-tech-and-state-of-the-art-radar-for-the-raf>, accessed 26 October 2022.

traditional air defence radar systems. The UK has a limited ability to counter these threats in the form of its Drone Dome system procured from Rafael Advanced Defense Systems in 2018, which is designed to target sUAS using soft-kill effects.[59] Additionally, the RAF has developed the ORCUS C-UAS system with Leonardo, which also employs soft-kill effects to defeat sUAS, although only four are expected to be delivered.[60] Complex threats such as cruise missiles and RPAS can be countered by the Army's Sky Sabre, which provides a detection range of 120 km and an engagement range of 25 km. It is claimed to be capable of hitting a target the size of a tennis ball travelling at the speed of sound.[61] Importantly, Sky Sabre is already capable of sharing data with Royal Navy and RAF assets in theatre via the Link 16 tactical datalink. Building on these foundations, an obvious priority candidate for investment in automated, real-time cross-domain connectivity in an IAMD context would be the RAF's new E-7A Wedgetail. It not only carries a high-resolution MESA radar that can contribute a look-down view of low-flying targets at far greater ranges than land-based radars but is also the standard Link 16 network gateway that would already be expected to interface between airborne and ground-based IAMD assets to build a common recognised air picture.[62]

In terms of 'shooters', the list of UK systems that can be considered for integration is currently quite short but in need of expansion to properly cover the wide range of potential threats. They include: Drone Dome and ORCUS for C-UAS effects; Starstreak and its vehicle-mounted high-velocity missile variant for short-range air defence; Sky Sabre for medium-range air defence; and the Royal Navy's ship-borne CAMM short-range and Aster 15 and 30 medium-/long-range missiles as part of the Type 45 Destroyer's Sea Viper air defence system. The ability to integrate high-end land-based capabilities such as Patriot PAC-3 or THAAD fielded by allies and partner nations would significantly increase potential effectiveness from both a 'shooter' and a 'sensor' point of view. In addition, any IAMD C2 network must incorporate high-fidelity data about friendly and civilian air operations in the airspace in question. Without an accurate, common and comprehensive recognised air picture it is very challenging to manage IAMD engagements safely and effectively. This requirement for deconfliction is another reason why E-7A, as the single best 'source of truth' within the air component about the aerial battlespace, would be a high-gain priority for any efforts to enhance IAMD effectiveness through JADC2-style connectivity.

The UK already has a somewhat developed capability that can be used to integrate sensors and effectors from multiple domains into a single IAMD. The SkyKeeper system from Lockheed Martin UK has been developed and refined to meet the Land Environment Air Picture Provision (LEAPP) capability requirement. A contract for its upgrade and service life extension was awarded in January 2022, which will improve the capabilities of the truck-mounted system and extend

59. Gareth Jennings, 'UK Signs for Drone Dome C-UAS System', *Janes Defence Weekly,* August 2018.
60. Richard Scott, 'UK Declares IOC Milestone for Synergia C-UAS Programme', *Janes International Defence Review,* September 2020.
61. Nicholas Fiorenza, 'British Army Receives Sky Sabre Air-Defence System', *Janes Defence Weekly,* December 2021.
62. Author interviews with E-7A Wedgetail crew and programme managers, RAF Waddington, 3 September 2020.

its service life to 2029.[63] SkyKeeper can integrate C-UAS, counter-rocket and mortar (C-RAM) and conventional air defence assets into a single recognised air picture for the operators of each system. In the future, the system is claimed to have significant growth potential for use as a battlespace C2 network, which would allow incoming threats to be identified and then allocated to the most suitable of a potentially wide variety of shooters such as C-RAM, Sky Sabre or C-UAS effectors. The current SkyKeeper version unveiled in 2019 can receive data feeds from most sensors and transmit target tracking data using common data links such as Link 16 from air assets and Link 11 from naval assets.[64] Lockheed Martin UK also claims that it can integrate the wide-area sense and detect sights of Ajax into the system, and can also integrate data from the F-35 Lightning II into the network if required.[65] This would be a significant expansion of the current LEAPP requirement set, but illustrates that there are already networking solutions in place with UK forces that could potentially be used as building blocks for increasing the real-time integration of sensors and potentially shooters from across the joint force in an IAMD context.

The vital importance of a high-fidelity common recognised air picture, and the multifaceted nature of the potential threats ranging from sUAS to salvoes of ballistic and cruise missiles that might face UK forces deployed in the Middle East make the priorities clear for early investments in JADC2-style cross-domain connectivity if IAMD is identified as a priority mission set for the future joint force. Significant increases in effectiveness and operational flexibility could be expected if existing network and C2 capabilities such as LEAPP that currently focus on connecting systems in a single domain could be expanded or linked to allow routine real-time data exchange with airborne sensor nodes such as the E-7A Wedgetail, F-35 and Crowsnest.

In reality, enabling full-spectrum IAMD coverage in a theatre as complex as the Middle East with capabilities currently in the UK inventory is unlikely to be possible. Seamless data sharing with allied ground-based capabilities such as Patriot PAC-3 and THAAD, as well as E-3A/F/G AWACS, would be necessary for deconfliction purposes but would also be a high priority due to the potential improvements to mutual sensor and shooter coverage.

63. Giles Ebbutt, 'British Army Upgrades Air Defence Capability', *Janes International Defence Review*, 26 January 2022.
64. Lockheed Martin, 'SkyKeeper Battlespace Management and Surveillance', 2019, <https://www.lockheedmartin.com/content/dam/lockheed-martin/uk/photo/products/skykeeper/skykeeper-datasheet.pdf>, accessed 26 October 2022.
65. Lockheed Martin UK, 'Battle Management Command, Control, Communications, Computers and Intelligence (BMC4I) for International Ground Based Air Defence (GBAD), Counter-Unmanned Aerial Systems (C-UAS) and Counter-Rocket, Artillery & Mortars (C-RAM)', Strategy White Paper, Version 3, p. 8.

IV. Vignette 3: Contested Freedom of Navigation Operations

THE ROYAL NAVY conducted a potentially opposed FONOP in June 2021 when HMS *Defender* sailed close to the Russian-occupied territory of Crimea in the Black Sea.[66] During this operation, *Defender* was harassed by Russian attack aircraft and gunboats and locked up by shore-based radar systems. The Royal Navy has also conducted FONOPs in the South China Sea close to artificial reefs built by the People's Liberation Army Navy to create illegal bastions of territory claims and area-denial system coverage.[67] The US Navy frequently conducts FONOPs, and they are seen by many NATO nations as a crucial politico-military tool to assert rights of access to the global commons against aggressive attempts to change borders or interdict sea lines of communication (SLOCs) by force.[68] With Russia committed to long-term confrontation with the West following the invasion of Ukraine in 2022, and China continuing its policies of attempting to change the status quo in both the South and East China seas, FONOPs are likely to remain an important tool for Western policymakers in the coming years.

However, the continuing development and proliferation of highly capable anti-ship missiles, submarines and more novel potential threats, such as swarms of sUAS and loitering munitions, will all ensure that even major surface action groups will face considerable danger during contested FONOPs at times of heightened tension. The US Navy's NIFC-CA architecture was developed precisely to enhance the survivability and lethality of naval task forces against high-end threats in a warfighting context by greatly improving the situational awareness and weapon effectiveness of the destroyers and aircraft against incoming threats.[69] At a less ambitious scale, it is worth examining where similar automated C2 and data-sharing functions between platforms would add most value for the UK in the context of FONOPs to defend access to contested SLOCs.

In a similar fashion to the preparatory stages of a SEAD/DEAD campaign, the first challenge in reducing the risk to a planned FONOP would be to try to detect, identify and then track the locations of the primary anti-ship threats in the contested maritime area. These will generally comprise coastal defence cruise missiles and the ground-based and possibly airborne radars that provide them with target information, as well as, potentially, missile-carrying small-boat

66. *BBC News,* 'HMS Defender: Russian Jets and Ships Shadow British Warship', 23 June 2021.
67. Tim Kelly, 'EXCLUSIVE – British Navy Warship Sails Near Beijing-Claimed South China Sea Islands – Sources', *Reuters*, 6 September 2018.
68. US Department of State, 'Freedom of Navigation Report Annual Release', 16 March 2021, <https://www.state.gov/freedom-of-navigation-report-annual-release>, accessed 30 May 2022.
69. O'Donoghue, McBirney and Persons, *Distributed Kill Chains*, pp. 41–50.

swarms, sUAS and in the case of larger powers possibly aircraft and submarines.[70] The nature of the threat in each geographic context will determine the scope and feasibility of mitigation and (if necessary) defence or deterrence measures to ensure a FONOP can be conducted successfully. However, in all cases a thorough understanding of enemy force dispositions and capabilities will be essential. Without such an understanding it is impossible to adequately protect a vessel conducting FONOPs during periods of high tension.

The second challenge will be to ensure that the ship or task group is provided with as many credible defence and/or retaliatory options as possible to deter hostile action by denial and/or punishment respectively. The third will be ensuring adequate connectivity between the ship conducting the FONOP, any supporting assets, and higher command levels who can provide rapid guidance on changing political circumstances, rules of engagement or orders throughout the period of vulnerability. This may be very difficult given that an adversary is likely to make liberal use of electronic warfare to interfere with the sensors and communications of the ship and any supporting elements, as such actions signal displeasure and create windows of kinetic vulnerability without the escalation implications of kinetic force.

Intelligence collection assets being used to ascertain the localised threat nature and location of key adversary capabilities are unlikely to be a major priority for integration if they will be conducting their missions well in advance of the FONOP in question. By its nature, the side conducting a FONOP generally has a significant degree of control over the timing of the passage through the contested waterway. Therefore, increasing the ability to pull information off ISR assets in real time through greatly increased connectivity is unlikely to provide more than a marginal increase in the efficiency of such preparatory activity compared to current practices. On the other hand, on the day of the FONOP itself, as the ship(s) in question sail towards the contested area, the ability to provide real-time connectivity between supporting ISR assets and the surface action group could add significant value. ISR assets such as P-8 Poseidon or RC-135W Rivet Joint may be able to detect any changes in the movement patterns and electronic emissions of adversary ground-based and maritime threat systems which could signal intent to conduct an attack. The faster this information can be relayed to the task group, the more effective any defensive and/or retaliatory measures can be made, and the greater the deterrent effect on an adversary.

During the FONOP transit itself, the ship's onboard radar systems would provide the primary source of defensive situational awareness, alongside the sensors of any embarked aviation – such as the MX15 sensor ball and SeaSpray AESA radar mounted on the Royal Navy's Wildcat HMA.2 helicopters.[71] The outputs from these sensors are theoretically already supposed to be well integrated via the Merlin Mk2 Crowsnest system which (if present) would also provide another

70. For more detail, see Watling, Bronk and Kaushal, 'A UK Joint Methodology for Assuring Theatre Access', pp. 7–22.

71. Leonardo, 'Seaspray: Surveilling the Seas', <https://uk.leonardo.com/en/news-and-stories-detail/-/detail/seaspray-surveilling-the-seas>, accessed 26 October 2022.

source of wide area radar surveillance to spot incoming threats.[72] These systems should, on their own, provide excellent radar coverage of both the aerial and surface threat picture during the hours of the transit itself. However, the interception of incoming missile threats would still rely on the radar picture from the ship's own sensors, which restricts the range at which such engagements can be conducted against sea-skimming threats. Sufficiently high-resolution/low-latency data sharing between airborne sensors (including F-35B, if in theatre) and the Type 45 to enable third-party missile guidance for Aster 30 could enable the ship to engage incoming threats beyond the radar horizon of its own sensor coverage. Against supersonic or hypersonic threats this would significantly enhance the odds of a successful defensive engagement and might also allow the launcher to be targeted rapidly with naval gunfire or TLAMs to prevent further hostile launches. This is precisely what NIFC-CA allows for the US Navy. However, the Royal Navy lacks a long-range missile capability comparable to the US SM-3 and SM-6 series, and aside from the F-35B's radar, the available airborne radars, which include Crowsnest and SeaSpray, may lack the technical capacity to provide missile guidance even if furnished with a sufficiently advanced ability to share data with both ship and missile in flight.

In principle, NIFC-CA or JADC2-style connectivity with airborne assets with high-end sensors offers the opportunity to enhance the guidance quality and effective range against sea-skimming or surface threats for the missiles carried by Royal Navy ships during a contested FONOP. This would not only offer the potential for lower risk against a given threat but also lower the political escalation risk during an actual engagement – since the number of munitions fired can prove the symbolic difference between a skirmish and a major confrontation at the geopolitical level.

The same can be said of both the precision and need for any retaliatory strikes against shore-based or maritime attackers. From an escalation-control point of view, the ability to intercept incoming threats is preferable to needing to go after the enemy launch site/vehicle, if possible, but where the latter is desirable or necessary, it is better to use fewer missiles to accomplish the task. Therefore, there are potential efficiencies to be gained from investment in JADC2-style connectivity for the likely force package the UK might deploy for a contested FONOP. However, they are marginal compared to the potentially dramatic combat effectiveness gains identified if initial investment were focused on priority areas in the context of an IAMD or SEAD/DEAD mission set.

72. Naval Technology, 'Royal Navy's First Merlin "Crowsnest" Helicopter Enters Service', 25 March 2021, <https://www.naval-technology.com/news/royal-navys-first-merlin-crowsnest-helicopter-enters-service/>, accessed 26 October 2022.

Conclusion

THIS PAPER IS intended to illustrate that a mission-facing approach could help UK policymakers identify the most relevant and potentially easiest capabilities across the force to concentrate on integrating in the early tranches of necessarily sequential MDI/JADO-type transformation efforts. Currently, there is a widely held quasi-philosophical position across the UK, the US and the armed forces of other nations that future wars will be decided by whichever side can best manage and share data seamlessly across all force elements.[73] The UK's Joint Concept Note 1/20 makes clear that the MoD is committed to a vision of the future force in which seamless cross-domain connectivity and integration is the norm. However, this high-level vision does not translate neatly into programme deliverables and prioritisation decisions within stretched budgets and limited programme management capacity. Even ambitious transformation projects have to start somewhere, and well-targeted initial investments in integrating a limited number of key capabilities across the domains could deliver significant enhancements to whole-force combat effectiveness.

The US Navy's approach, with its NIFC-CA programme model, offers a potentially useful model for the UK to pursue JADC2-type connectivity across the force in more manageable and easily defined blocks. Rather than attempting root-and-branch change across the whole force simultaneously, the US Navy created an unprecedented level of real-time integration between surface warships, airborne assets and weapons in flight by focusing initially on connecting the core platforms essential to a given mission, and then building on the progress made once initial investment had yielded real-world results.

For the UK, the lessons are clear:

- The MoD, and Strategic Command as the integrating authority, need to prioritise early investment in real-time data sharing and combined C2 capabilities between the platforms and systems likely to deliver the greatest improvement in combat effectiveness for the force in specific key mission sets.
- Mission-specific analysis can be used to identify which force elements would make most sense to concentrate early investment tranches on, by providing a framework through which to examine if and where advanced cross-domain data sharing and third-party targeting capabilities between specific platforms would add the most tangible value in an operational context.
- In general, even a superficial analysis of kinetic mission sets such as the ones provided in this paper suggests that enhanced data sharing and third-party targeting capabilities linking key sensor assets, such as F-35, E-7A and SOF teams, with long-range precision

73. See Watling, 'More Sensors than Sense'.

fires and IAMD capabilities like GMLRS and Type 45, could greatly enhance combat effectiveness in challenging scenarios where the current force structure would struggle.

It is also worth noting that analysis of some mission sets may suggest that for many force elements the potential combat effectiveness gains from currently deliverable JADC2-style integration would offer significantly less in terms of return on investment in the near term than other options for that funding. This should not be seen as a bad thing since careful prioritisation is undoubtedly necessary across defence, given limited budgets, personnel headroom and programme management capacity. Therefore, choosing which assets across the joint force do not yet need to be connected by an all-singing, all-dancing network and automated C2 solution is perhaps just as important as identifying what assets ought to be connected with all possible speed.

The future of defence may eventually include something akin to the vision laid out in Joint Concept Note 1/20 and other top-level strategy documents; a completely connected joint force, fully integrated in real-time not only across all domains but also across government departments. However, there is a huge gulf between this ambition and the current state of the joint force. With limited resources and a host of urgent force modernisation and regeneration requirements, the UK's short-to-medium investments in MDI/JADO-type capabilities will need to be incremental and focus on seamlessly connecting and integrating the assets that will see the greatest combat effectiveness increase from the capability to share real-time situational awareness and conduct joint engagements.

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